

NOTES ON NONRESPONSE WEIGHTING AND MULTILEVEL REGRESSION POST-STRATIFICATION

DOUGLAS RIVERS

1. INTRODUCTION

Most samples used today for the study of public opinion and political behavior do not come close to approximating probability samples, due to high levels of non-response, panel attrition, or opt-in designs. Sample observations are usually weighted to adjust for non-response. These weights are *not* the reciprocal of sample selection probabilities (which are not known, even approximately).

The basic idea behind weighting is fairly simple: we divide the sample into subsets (“poststrata”) whose frequency in the population is known from a census or other reliable source. Observations in each poststrata receive the weight

$$\text{weight} = \frac{\text{Population proportion in poststratum}}{\text{Sample proportion in poststratum}}$$

The weighted sample proportion in each poststratum is equal to the population proportion in that poststratum.

Various problems arise with post-stratification weighting:

- If there are no sample observations in a poststratum, then it is impossible to calculate the weight.
- When the weights are large for some poststrata, then the resulting estimator will tend to have a large variance.

To avoid these problems, it is common to “collapse cells” (*i.e.* combine poststrata) (so there are no empty cells) and to “trim” the weights (reduce the size of large weights). These are attempts to trade off some bias for a reduction in the variance. Neither approach is very attractive. There is little theoretical justification for either. The adjustments are crude and usually arbitrary.

Hierarchical Bayesian methods, in the form of *Multilevel Regression and Post-stratification* (a.k.a. “Mr. P”) will be shown to be a better alternative.

2. POST-STRATIFICATION

We divide the sample into H post-strata or weighting classes, *e.g.* men and women under 25 and over 25 (“quads”). Suppose poststratum h has N_h units in the population, of which n_h are selected for the sample. However, only r_h of the selected units respond. (With some abuse to notation, we will use r_h to denote both the set of responding units $r_h \subset s_h \subset U_h$ as well as the number of

responding units.) Let $\bar{y}_h$ denote the sample mean in the h th poststratum *among the responding units* and μ_h the corresponding population mean. The population mean and total are

$$t_U = \sum_{h=1}^H N_h \bar{y}_{U_h} \quad \bar{y}_U = \sum_{h=1}^H \frac{N_h}{N} \bar{y}_{U_h}$$

The poststratified estimator is

$$\bar{y}_{PS} = \sum_{h=1}^H \frac{N_h}{N} \bar{y}_{r_h} = \frac{\sum_{i \in r_h} w_i y_i}{\sum_{i=1}^r w_i}$$

where $w_i = N_h/r_h$ if i is in poststratum h and $r = r_1 + \dots + r_H$ is the total number of responding units.

Next, we derive the variance and mean square error of the post-stratification estimator under the MAR assumption, where the covariate X_i indicates the poststratum each unit belongs to. However, under MAR, the responding units in each poststratum are a SRSWOR from the selected units in the poststratum (conditional upon $\mathbf{r} = (r_1, \dots, r_H)$) and the poststratum means are independent. Therefore,

$$\mathbb{V}(\bar{y}_{PS}|\mathbf{r}) = \sum_{h=1}^H \left(\frac{N_h}{N} \right)^2 (1 - r_h/N_h) \frac{\sigma_{yh}^2}{r_h}$$

where

$$\sigma_{yh}^2 = \mathbb{V}(Y_i|i \in U_h)$$

The poststratified estimator is unbiased (conditional upon $r_h > 0$ for each h). Oh and Scheuren derive a similar result (with s_{yU_h} in place of σ_{yh} using the assumption of a uniform response mechanism within subpopulations using quasi-randomization).

Problems in post-stratification weighting:

- Bias. The cell estimates are ratios and subject to the usual bias for ratio estimators. This is not important if the cell sizes are large, but becomes a serious issue with small cell sizes. The usual recommendation is to have at least 30 sample observations per cell.
- Curse of dimensionality. Attempting to weight on a large number of attributes increases the number of cells at an exponential rate, leading to cells with few or no observations. A full age $\times$ gender $\times$ race $\times$ education cross-classification results in roughly $4 \times 2 \times 3 \times 4 = 96$ cells, or an average of less than ten observations per cell for a sample of 1,000. In fact, some of these cells are almost guaranteed to be empty (*e.g.*, 65+ female Hispanic with a post-graduate degree).
- Population targets for cross-classifications. Frequently the weighting variables come from different sources, *e.g.* demographic data from the census and attitudinal variables (like party identification) from opinion surveys, so we only have data on the marginal distributions of the weighting variables, not their cross-classifications.

3. RAKING

When there is nonresponse, sample marginals often differ substantially from known population marginals on demographics and other variables. The method of *raking* (a.k.a. *rim weighting*) was

introduced by Deming and Stephan (1940) to “adjust” for nonresponse. The goal of raking is to produce a set of weights which force the marginal distribution of the sample variables to match known population marginals. The idea of raking is simple and easy to implement:

- (1) Given a set of weights, w_i , form post-strata using a variable whose marginal distribution is known (*e.g.*, from the Census). Multiply the weights in each post-stratum by the ratio of the known population marginal to the sample marginal. This guarantees that the sample marginals with the adjusted weights match the population marginals on this variables.
- (2) Repeat (1) for the other variables. At each stage, the chosen sample marginal distribution will match the population marginal (though there will be differences for the other variables, including those previously adjusted).
- (3) Iterate until convergence.

This is called *iterative proportional fitting*. We will show that this approximately solves a chi-square minimization method (which, interestingly, does not require iterative solution, but is rarely used).

We discuss two methods of raking. The original method, proposed by Deming and Stephan (1940) uses a minimum chi-square estimator. Ireland and Kullback (1968) propose an asymptotically equivalent method using minimum discrepancy (also called the likelihood ratio chi-square), which is usually computed using iterative proportional fitting. We illustrate the methods using two-way cross-classifications, though both methods extend to higher dimensional tables. These methods are equivalent to a log-linear model for the cross-classification with no interaction. The discussion here follows Bishop, Fienberg and Holland, *Discrete Multivariate Analysis* (MIT Press, 1975).

Raking does not solve the nonresponse problem. If nonresponse is MCAR, raking will improve the efficiency of estimates (but there is no bias to remove). However, if nonresponse is MAR and the covariates are used to rake, raking is inconsistent (if the covariates do not obey the non-interaction condition). This alone should cause us to reject its use in most applications, though it remains popular.

4. COMPUTATION

Before discussing the theory of raking, we provide an example from Lohr.

```
> # Raking example from Lohr, p. 344
> df <- data.frame(gender=rep(c("Female", "Male"), c(1620, 1380)),
+   race=factor(rep(c(1:5, 1:5), c(300, 1200, 60, 30, 30, 150, 1080, 90, 30, 30))))
> df$race <- ordered(df$race, labels=c("Black", "White", "Asian", "Native American", "Other"))
> with(df, table(race, gender))
```

	gender	
race	Female	Male
Black	300	150
White	1200	1080
Asian	60	90
Native American	30	30
Other	30	30

```
> with(df, round(100 * prop.table(table(race, gender), 1)))
      gender
race      Female Male
Black      67    33
White      53    47
Asian      40    60
Native American 50    50
Other      50    50
>
```

The sample has too many females, especially among Blacks. Suppose, based on the census, we thought that the sample counts for a sample of size $n = 3000$ should be as follows:

```
> library(survey)
> pop.gender <- data.frame(gender=c("Female","Male"), Freq=c(1510,1490))
> pop.race <- data.frame(race=ordered(c("Black","White","Asian","Native American",
+   Freq=c(600,2120,150,100,30))
```

(In most cases, the census provides percentages, which would need to be multiplied by the sample size ($nrow(df)$). We can now use the survey library to post stratify by, say, gender:

```
> library(survey)
> pop.gender <- data.frame(gender=c("Female","Male"), Freq=c(1510,1490))
> pop.race <- data.frame(race=ordered(c("Black","White","Asian","Native American",
+   Freq=c(600,2120,150,100,30))
>
> library(survey)
> design0 <- svydesign(id=~1, data=df)
Warning message:
In svydesign.default(id = ~1, data = df) :
  No weights or probabilities supplied, assuming equal probability
> svytable(~gender+race, design0)
      race
gender Black White Asian Native American Other
Female   300  1200    60             30    30
Male    150  1080    90             30    30
> design1 <- postStratify(design0, ~gender, pop.gender) # post-stratify by gender
> svytable(~gender+race, design1)
      race
gender Black White Asian Native American Other
Female 279.62963 1118.51852 55.92593      27.96296 27.96296
Male   161.95652 1166.08696 97.17391      32.39130 32.39130
> rowSums(svytable(~gender+race, design1))
Female   Male
  1510    1490
```

This fixed the gender distribution, but now the race distribution is off. Let's try post stratifying on race:

```
> design2 <- postStratify(design1, ~race, pop.race) # post-stratify by race
```

```

> svytable(~gender+race, design2)
      race
gender  Black      White      Asian Native American      Other
Female  379.94348 1037.92943  54.79358          46.33138  13.89941
Male    220.05652 1082.07057  95.20642          53.66862  16.10059
> colSums(svytable(~gender+race, design2))
      Black      White      Asian Native American
      600      2120      150          100
      Other
      30

```

This fixed the race distribution, but now the gender distribution is off. Raking iterates this process until both constraints are satisfied:

```

> # rake (keep post-stratifying until convergence)
> design3 <- rake(design0, sample.margins=list(~gender, ~race),
+   population.margins=list(pop.gender, pop.race),
+   control=list(epsilon=.00001, maxit=100, verbose=TRUE))
      race
gender  Black White Asian Native American Other
Female   300  1200   60          30   30
Male     150  1080   90          30   30
      race
gender  Black      White      Asian Native American      Other
Female  379.94348 1037.92943  54.79358          46.33138  13.89941
Male    220.05652 1082.07057  95.20642          53.66862  16.10059
      race
gender  Black      White      Asian Native American      Other
Female  375.67121 1021.75848  53.73602          45.57298  13.67189
Male    224.32879 1098.24152  96.26398          54.42702  16.32811
      race
gender  Black      White      Asian Native American      Other
Female  375.59431 1021.46870  53.71714          45.55940  13.66782
Male    224.40569 1098.53130  96.28286          54.44060  16.33218
      race
gender  Black      White      Asian Native American      Other
Female  375.59293 1021.46350  53.71680          45.55916  13.66775
Male    224.40707 1098.53650  96.28320          54.44084  16.33225
> svytable(~gender+race, design3)
      race
gender  Black      White      Asian Native American      Other
Female  375.59293 1021.46350  53.71680          45.55916  13.66775
Male    224.40707 1098.53650  96.28320          54.44084  16.33225
> rowSums(svytable(~gender+race, design3))
Female  Male
  1510   1490
> colSums(svytable(~gender+race, design3))
      Black      White      Asian Native American
      600      2120      150          100

```

5. MINIMUM CHI-SQUARE ESTIMATION

Suppose we observe an $I \times J$ table of counts $\{n_{ij}\}$ with marginals $\{n_{i+}\}$ and $\{n_{+j}\}$. We wish to obtain weights $\{w_{ij}\}$ such that the adjusted counts

$$\hat{n}_{ij} = w_{ij}n_{ij}$$

have given marginals

$$\hat{n}_{i+} = \sum_{j=1}^J \hat{n}_{ij} = n_{i+}^0 \quad (i = 1, \dots, I)$$

and

$$\hat{n}_{+j} = \sum_{i=1}^I \hat{n}_{ij} = n_{+j}^0 \quad (j = 1, \dots, J).$$

Both marginals sum to sample size:

$$\sum_{i=1}^I n_{i+}^0 = \sum_{j=1}^J n_{+j}^0 = n.$$

Subject to these restrictions, we choose the adjusted counts to minimize the chi-squared statistic,

$$\chi^2(\hat{n}_{ij}) = \sum_{i=1}^I \sum_{j=1}^J \frac{(\hat{n}_{ij} - n_{ij})^2}{n_{ij}}.$$

We form the Lagrangean

$$\mathcal{L} = \frac{1}{2} \sum_{i=1}^I \sum_{j=1}^J \frac{(\hat{n}_{ij} - n_{ij})^2}{n_{ij}} + \sum_{i=1}^I \lambda_i (n_{i+}^0 - \hat{n}_{i+}) + \sum_{j=1}^J \gamma_j (n_{+j}^0 - \hat{n}_{+j})$$

where $\lambda_1, \dots, \lambda_I$ and $\gamma_1, \dots, \gamma_J$ are Lagrange multipliers. This gives the first order conditions

$$\frac{n_{ij} - \hat{n}_{ij}}{n_{ij}} + \lambda_i + \gamma_j = 0 \quad (i, j = 1, \dots, J)$$

$$\hat{n}_{i+} = n_{i+}^0 \quad (i = 1, \dots, J)$$

$$\hat{n}_{+j} = n_{+j}^0 \quad (j = 1, \dots, J)$$

The first condition implies that

$$\hat{n}_{ij} = (1 + \lambda_i + \gamma_j)n_{ij} \quad (i = 1, \dots, I; j = 1, \dots, J)$$

Thus, the weights $w_{ij} = 1 + \lambda_i + \gamma_j$. That is, the weights include a row adjustment (λ_i) and a column adjustment (γ_j) which enter additively. With more variables, there is an additional adjustment per variable. This obviously is feasible only if $n_{ij} > 0$ for all i and j .

Next, we need to solve for $\lambda_1, \dots, \lambda_I$ and $\gamma_1, \dots, \gamma_J$. The idea is that the first order condition

$$n_{ij} - \hat{n}_{ij} + \lambda_i n_{i+} + \gamma_j n_{+j} = 0$$

can be summed over either i and j to obtain

$$\begin{aligned}\sum_{j=1}^J n_{ij} - \sum_{j=1}^J \hat{n}_{ij} + \sum_{j=1}^J n_{ij} \gamma_j + \lambda_i \sum_{j=1}^J n_{ij} &= 0 & (i = 1, \dots, I) \\ \sum_{i=1}^I n_{ij} - \sum_{i=1}^I \hat{n}_{ij} + \sum_{i=1}^I n_{ij} \lambda_i + \gamma_j \sum_{i=1}^I n_{ij} &= 0 & (j = 1, \dots, J)\end{aligned}$$

or, equivalently,

$$\begin{aligned}\sum_{j=1}^J n_{ij} \gamma_j + n_{i+} \lambda_i &= n_{i+}^0 - n_{i+} & (i = 1, \dots, I) \\ n_{+j} \gamma_j + \sum_{i=1}^I n_{ij} \lambda_i &= n_{+j}^0 - n_{+j} & (j = 1, \dots, J)\end{aligned}$$

In matrix form, we have the set of linear equations,

$$\begin{pmatrix} n_{1+} & \cdots & 0 & n_{11} & \cdots & n_{1J} \\ \vdots & \ddots & \vdots & \vdots & & \vdots \\ 0 & \cdots & n_{I+} & n_{I1} & \cdots & n_{IJ} \\ n_{11} & \cdots & n_{I1} & n_{+1} & \cdots & 0 \\ \vdots & & \vdots & \vdots & \ddots & \vdots \\ n_{1J} & \cdots & n_{IJ} & 0 & \cdots & n_{+J} \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_I \\ \gamma_1 \\ \vdots \\ \gamma_J \end{pmatrix} = \begin{pmatrix} n_{1+}^0 - n_{1+} \\ \vdots \\ n_{I+}^0 - n_{I+} \\ n_{+1}^0 - n_{+1} \\ \vdots \\ n_{+J}^0 - n_{+J} \end{pmatrix}$$

These equations are easily seen to be linearly dependent, but it can be shown that the matrix on the left has rank $I + J - 1$, so a unique solution can be obtained by eliminating the last row and column of the matrix, solving for $\lambda_1, \dots, \lambda_I, \gamma_1, \dots, \gamma_{J-1}$, and then solve for λ_J :

$$\lambda_J = -\frac{1}{n_{+J}} \left(\sum_{i=1}^I \lambda_i n_{i+} + \sum_{j=1}^{J-1} \gamma_j n_{+j} \right).$$

There is no guarantee that the solution to this system is strictly positive, so it could produce negative weights. It is asymptotically equivalent to the method described in the next section provided all row and column marginals are strictly positive. The alternative method always produces positive weights and is the algorithm used by *R*. It does not, however, have a closed form solution.

6. MINIMUM DISCREPANCY ESTIMATION

The iterative proportional fitting (IPF) algorithm does not actually result in the minimum chi-square estimates, though it usually differs only slightly. Ireland and Kullback (1968) show that the IPF algorithm minimizes the KLIC. Let $p_{ij} = n_{ij}/n$ denote the sample proportion in cell (i, j) and $\hat{p}_{ij} = \hat{n}_{ij}/n$ the adjusted estimate. The KLIC is

$$\text{KLIC}(\hat{p}_{ij}) = \sum_{i=1}^I \sum_{j=1}^J \hat{p}_{ij} \log(\hat{p}_{ij}/p_{ij})$$

or, in terms of cell counts,

$$\text{KLIC}(\hat{n}_{ij}) = \frac{1}{n} \sum_{i=1}^I \sum_{j=1}^J \hat{n}_{ij} \log(\hat{n}_{ij}/n_{ij})$$

A similar development as in the proceeding section shows that the solution satisfies

$$\hat{n}_{ij} = \lambda_i \gamma_j n_{ij}$$

so the weights are given by

$$w_{ij} = \lambda_i \gamma_j.$$

These can be obtained by repeated rounds of raking. Ireland and Kullback show that the rate of convergence is geometric.

7. WEIGHT TRIMMING

When you do poststratification weighting for nonresponse, you are likely to encounter a *bias–variance tradeoff*.

- If you use many poststrata, the sample fraction in some poststrata can be very small (or, worse, zero), and the weights get very large and the variance of the estimates becomes large.
- So you “trim” the weights (so they aren’t so large) or “collapse” some cells (so there aren’t so few observations in small poststrata), which causes bias.

You either tolerate large variance or risk large bias.

Trimming is accomplished by changing the weights that fall outside of some interval $(w_{\min}, w_{\max})$,

$$w_i^* = \begin{cases} w_{\min} & \text{if } w_i \leq w_{\min} \\ w_i & \text{if } w_{\min} < w_i < w_{\max} \\ w_{\max} & \text{if } w_i \geq w_{\max} \end{cases}$$

(Often trimming is limited to large weights, so $w_{\min} = 0$). Using the trimmed weights,

$$\bar{y}_{\text{trimmed}} = \frac{\sum_{i \in s} w_i^* y_i}{\sum_{i \in s} w_i^*} = \sum_{h=1}^H \frac{\tilde{N}_h}{N} \hat{\mu}_h$$

where $\tilde{N}_h$ is the implied estimate of N_h using the trimmed weight. The untrimmed poststratification weight estimates N_h exactly (by construction). Since $\mathbb{E}(\hat{\mu}_h) = \mu_h$, trimming causes bias:

$$\text{Bias}(\hat{\mu}_{\text{trimmed}}) = \mathbb{E}(\hat{\mu}_{\text{trimmed}}) - \mu = \sum_{h=1}^H \mathbb{E} \left(\frac{\tilde{N}_h - N_h}{N} \mu_h \right) + \mathbb{E} \left(\frac{\tilde{N}_h}{N} (\hat{\mu} - \mu) \right)$$

This doesn’t make much sense, since the problem is the poor estimate of μ_h (caused by few sample observations), not the (correct) population proportion in the poststratum.

Collapsing is crude, but, if done correctly, approximates the right solution. If we combine two cells that have similar values of μ_h , this amounts to “borrowing strength” from similar cells. The

problem is that collapsing is *ad hoc* and doesn't use the data efficiently. If x_i is observed for every population unit, we could instead use the estimate

$$\hat{\tau} = \sum_{k \in \omega} y_k + \sum_{i \notin \omega} \hat{y}_i$$

where $\hat{y}_i = \hat{y}_i(x_i)$ is the predicted value of y_i for the unselected y_i . Collapsing amounts to using a weighted average of the sample y_k 's for all of the units in the combined poststrata.

In Bayesian inference, we treat the knowns ($y_k, k \in \omega$ and $x_1, \dots, x_N$) as known, and the unknowns ($y_k, k \notin \omega$) as unknowns. The combination of a prior and a model for the data produce optimal predictions for the unknowns using Bayes' Theorem.

8. BAYESIAN ANALYSIS OF SAMPLING FROM A NORMAL SUPERPOPULATION

We will assume that the population measurements $y_1, \dots, y_N$ are realizations of a superpopulation process $Y_1, \dots, Y_N$, where the Y_i 's are i.i.d. $N(\mu, \sigma^2)$. We have a random sample $y_{i_1}, \dots, y_{i_n}$ of fixed size n . (It makes no difference how the sample was selected so long as sampling is noninformative, *i.e.*, the selection probabilities do not depend on the y_i 's.)

8.1. Conjugate Normal Prior. We do not have any real prior information about the superpopulation parameters μ and σ^2 , so we adopt a “noninformative” prior. However, we do know that $\sigma^2 > 0$, so it's easier to redefine the parameters to be $(\mu, \log \sigma^2)$ (so the parameter space becomes all of $\mathbb{R}^2$) and to adopt an improper uniform prior over $\mathbb{R}^2$. That is,

$$p_{\mu, \log \sigma^2}(\mu, \log \sigma^2) \propto 1$$

so

$$p_{\mu, \sigma^2}(\mu, \sigma^2) = p_{\mu, \log \sigma^2}(\mu, \log \sigma^2) \left| \frac{d \log \sigma^2}{d \sigma^2} \right| \propto \frac{1}{\sigma^2}.$$

8.2. Normal Likelihood. The likelihood for the sample y_k ($k \in \omega$) is

$$\begin{aligned} p(y_k, k \in \omega) &= \prod_{k \in \omega} p(y_k | \mu, \sigma^2) = \prod_{k \in \omega} \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y_k - \mu)^2}{2\sigma^2} \right\} \\ &= (2\pi\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{k \in \omega} (y_k - \mu)^2 \right\} \\ &= (2\pi\sigma^2)^{-n/2} \exp \left\{ -\frac{(n-1)\bar{\sigma}^2 + n(\bar{y} - \mu)^2}{2\sigma^2} \right\} \end{aligned}$$

Alternatively, we could (as in design-based inference) treat the y_k 's as parameters with an exchangeable prior distribution. This leads to an equivalent hierarchical model.

8.3. Normal-Inverse χ^2 Posterior. The posterior is proportional to the product of the prior and the likelihood, so

$$\begin{aligned} p(\mu, \sigma^2 | y_k, k \in \omega) &\propto \frac{1}{\sigma^2} (2\pi\sigma^2)^{-n/2} \exp \left\{ -\frac{(n-1)\tilde{\sigma}^2 + n(\bar{y} - \mu)^2}{2\sigma^2} \right\} \\ &\propto \frac{1}{\sqrt{2\pi\sigma^2/n}} \exp \left\{ -\frac{(\mu - \bar{y})^2}{2\sigma^2/n} \right\} (\sigma^2)^{-(n+1)/2} \exp \{ (n-1)\tilde{\sigma}^2/\sigma^2 \}. \end{aligned}$$

The marginal posterior for σ^2 given y_k ($k \in \omega$) is

$$\begin{aligned} p(\sigma^2 | y_k, k \in \omega) &= \int p(\mu, \sigma^2 | y_k, k \in \omega) d\mu \\ &\propto (\sigma^2)^{-(n+1)/2} \exp \{ (n-1)\tilde{\sigma}^2/\sigma^2 \} \int \frac{1}{\sqrt{2\pi\sigma^2/n}} \exp \left\{ -\frac{(\mu - \bar{y})^2}{2\sigma^2/n} \right\} d\mu \\ &= (\sigma^2)^{-(n+1)/2} \exp \left\{ -\frac{(n-1)\tilde{\sigma}^2}{2\sigma^2} \right\} \end{aligned}$$

This is proportional to the pdf of a scaled inverse chi-square random variable where the degrees of freedom are $n-1$ and the scale factor is $\tilde{\sigma}^2$.

The conditional posterior for μ given the sample y_k ($k \in \omega$) and σ^2 is evidently

$$p(\mu | \sigma^2, y_k, k \in \omega) = \frac{1}{\sqrt{2\pi\sigma^2/n}} \exp \left\{ -\frac{(\mu - \bar{y})^2}{2\sigma^2/n} \right\}$$

which is a $N(\bar{y}, \sigma^2/n)$ distribution.

8.4. The Predictive Distribution for the Population Total. The population total is

$$\tau = \sum_{k \in \omega} y_k + \sum_{k \notin \omega} y_k.$$

The first term is known from the sample. The distribution of the second term, conditional upon μ , σ^2 , and y_k ($k \in \omega$) is normal with mean $(N-n)\mu$ and variance $(N-n)\sigma^2$ (since $y_1, \dots, y_N$ are independent $N(\mu, \sigma^2)$ conditional upon μ and σ^2). It follows that the conditional distribution of τ given μ , σ^2 , and y_k ($k \in \omega$) is also normal, but with mean $n\bar{y} + (N-n)\mu$ and variance $(N-n)\sigma^2$, or

$$p(\tau | \mu, \sigma^2, y_k, k \in \omega) = \frac{1}{\sqrt{2\pi(N-n)\sigma^2}} \exp \left\{ -\frac{(\tau - n\bar{y} - (N-n)\mu)^2}{2(N-n)\sigma^2} \right\}.$$

The predictive distribution of τ given y_k ($k \in \omega$) is obtained by integrating out μ and σ^2 using their posterior distributions:

$$\begin{aligned} p(\tau | y_k, k \in \omega) &= \int \int p(\tau | \mu, \sigma^2, y_k, k \in \omega) p(\mu, \sigma^2 | y_k, k \in \omega) d\mu d\sigma^2 \\ &\propto \int (\sigma^2)^{-(n-1)/2} \exp \left\{ -\frac{(n-1)\tilde{\sigma}^2}{2\sigma^2} \right\} \times \\ &\quad \int \exp \left\{ -\frac{(\tau - n\bar{y} - (N-n)\mu)^2}{2(N-n)\sigma^2} \right\} \exp \left\{ -\frac{(\mu - \bar{y})^2}{2\sigma^2/n} \right\} d\mu d\sigma^2 \end{aligned}$$

After some extremely tedious manipulation, this can be simplified to

$$p(\tau|y_k, k \in \omega) \propto \left(1 + \frac{1}{n-1} \left(\frac{\tau - N\bar{y}}{\sqrt{N^2(1-n/N)\tilde{\sigma}^2/n}} \right)^2 \right)^{-[(n-1)+1]/2}$$

which is the pdf of a Student t distribution with $n-1$ degrees of freedom.

It is very difficult in the design-based theory to justify the use of a t distribution or degrees of freedom (all the results are just asymptotic normal), but the conventional frequentist practice of using the t_{n-1} distribution falls out of the Bayesian analysis!

9. HIERARCHICAL PRIORS

Superpopulation models may seem somewhat artificial. An alternative approach is to treat the N population measurements $y_1, \dots, y_N$ as parameters and to specify a prior over the N dimensional parameter space. This is the approach taken by Ericson, “Subjective Bayesian Models in Sampling Finite Populations,” *Journal of the Royal Statistical Society, Series B* (1969).

Under this setup, the likelihood is noninformative, except that the particular data values were observed. The parameter space for $(y_1, \dots, y_N)$ is $\mathbb{R}^N$ and

$$L(y'_1, \dots, y'_N) = \begin{cases} p(\omega) & \text{if } y'_k = y_k \text{ for } k \in \omega \\ 0 & \text{otherwise} \end{cases}$$

and $p(\omega)$ is the probability that the sample ω is selected. For given ω , any estimate consistent with the observed sample values gives the same value of the likelihood.

Choosing a prior in a high dimensional parameter space can be tricky. We might be tempted to assume that the y_i 's are independent $N(\mu, \sigma^2)$. In this case,

$$\mathbb{E}(\tau|y_k, k \in \omega) = \sum_{k \in \omega} y_k + \sum_{i \notin \omega} \mathbb{E}(y_i|y_k, k \in \omega) = n\bar{y} + (N-n)\mu$$

since, if $i \notin \omega$, then y_i is independent of y_k ($k \in \omega$). The problem is that the sample y_k 's aren't related in any way to the out-of-sample y_i 's.

Ericson suggested an exchangeable prior distribution for $(y_1, \dots, y_N)$,

$$p(y_1, \dots, y_N) = \int \prod_{i=1}^N p(y_i|\theta) d\theta$$

where the θ is a “superparameter.” Ericson treats the case where $p(y_i|\theta)$ is a $N(\mu, \sigma^2)$ distribution (where σ^2 is known) and the (hyper) prior for μ is diffuse, then

$$\mathbb{E}(\tau|y_k, k \in \omega) = N\bar{y}$$

and

$$\mathbb{V}(\tau|y_k, k \in \omega) = N^2 \left(1 - \frac{n}{N} \right) \frac{\sigma^2}{n}$$

which should look familiar.

Incidentally, none of these results depend upon random sampling (or, for that matter, any particular sampling scheme).

10. MULTILEVEL REGRESSION AND POST-STRATIFICATION

We are looking for a compromise between full poststratification and a classical regression model.

We shall assume throughout simple random sampling from a large population, so the base weights are all equal to one and the finite population corrections can be ignored.

10.1. Poststratification Weights. The poststratification estimator is

$$\hat{\mu}_{\text{PS}} = \sum_{h=1}^H \frac{N_h}{N} \hat{\mu}_h = \frac{1}{n} \sum_{k \in \omega} w_k y_k$$

where the weights are the

$$w_k = \frac{N_h/N}{n_h/n} = \frac{nN_h}{n_h N_h} \quad \text{for } k \in \omega_h$$

Note that the weights have been scaled so that

$$\sum_{k \in \omega} w_k = \sum_{h=1}^H \sum_{k \in \omega_h} w_k = \sum_{h=1}^H n_h \left(\frac{nN_h}{n_h N_h} \right) = \frac{n}{N} \sum_{h=1}^H N_h = n.$$

11. REGRESSION WEIGHTS

The poststratification estimator *is* a regression estimator. Let

$$x_{ih} = \begin{cases} 1 & \text{if } i \in U_h \\ 0 & \text{otherwise} \end{cases}$$

and let $\mathbf{X} = (x_{kh})$ be the $n \times H$ matrix containing the poststratum indicators for the n sample units. The least squares regression,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \begin{pmatrix} n_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & n_H \end{pmatrix}^{-1} \begin{pmatrix} \sum_{k \in \omega_1} y_k \\ \vdots \\ \sum_{k \in \omega_H} y_k \end{pmatrix} = \begin{pmatrix} \hat{\mu}_1 \\ \vdots \\ \hat{\mu}_H \end{pmatrix}$$

Let $\mathbf{N} = (N_1, \dots, N_H)^T$ denote the vector of poststratum sizes and $\mathbf{X}_{\text{pop}} = \mathbf{I}_H$ (the reason for this notation will become apparent later), so

$$\hat{\mu}_{\text{PS}} = \frac{1}{N} \mathbf{N}^T \mathbf{X}_{\text{pop}} \hat{\boldsymbol{\beta}} = \frac{1}{n} \left(\frac{n}{N} \right) \mathbf{N}^T \mathbf{X}_{\text{pop}} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \frac{1}{n} \mathbf{w}^T \mathbf{y} = \frac{1}{n} \sum_{k \in \omega} w_k y_k$$

where

$$\mathbf{w} = \frac{n}{N} \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}_{\text{pop}}^T \mathbf{N}$$

is the vector of weights $(w_1, \dots, w_n)^T$.

The result above applies to any regression that contains a constant (or, equivalently, a constant vector is contained in the span of $\mathbf{X}$). Rather than including indicators for all poststratification

cells, we might entertain a simpler model. For example, we might assume a linear functional form for one of the poststratifying variables (rather than dividing each variable into categories) or exclude certain interactions among the poststratifying variables. In this case, the regressor matrix $\mathbf{X}$ will be $n \times p$ where $p < H$. However, we will assume that the covariates are constant within each poststratification cell, so $\mathbf{X}_{\text{pop}}$ is an $H \times p$ matrix giving the value of the covariates in poststratification cell h .

The least squares regression coefficient $\hat{\beta}$ is still

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

but its dimension is now $p \times 1$. The poststratification estimator using classical regression is

$$\begin{aligned} \hat{\mu}_{\text{PS}} &= \frac{1}{N} \mathbf{N}^T \mathbf{X}_{\text{pop}} \hat{\beta} \\ &= \frac{1}{n} \left(\frac{n}{N} \right) \mathbf{N}^T \mathbf{X}_{\text{pop}} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \\ &= \frac{1}{n} \mathbf{w}^T \mathbf{y} = \frac{1}{n} \sum_{k \in \omega} w_k y_k \end{aligned}$$

where the weights

$$\mathbf{w} = \frac{n}{N} \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}_{\text{pop}}^T \mathbf{N}$$

are normalized to sum to sample size.

11.1. Hierarchical Regression. Next, we consider a Hierarchical Bayes (HB) regression model where

$$\mathbf{y} \sim N(\mathbf{X}\beta, \Sigma_y)$$

and the prior for β is

$$\beta \sim N_p(\mathbf{0}_p, \Sigma_\beta).$$

The posterior mean of β (when Σ_β is known) is

$$\hat{\beta} = (\mathbf{X}^T \Sigma_y^{-1} \mathbf{X} + \Sigma_\beta^{-1})^{-1} \mathbf{X}^T \Sigma_y^{-1} \mathbf{y}$$

The poststratified estimator is now

$$\hat{\mu}_{\text{PS}} = \frac{1}{N} \mathbf{N}^T \mathbf{X}_{\text{pop}} (\mathbf{X}^T \Sigma_y^{-1} \mathbf{X} + \Sigma_\beta^{-1})^{-1} \mathbf{X}^T \Sigma_y^{-1} \mathbf{y} = \frac{1}{n} \mathbf{w}_y^T \mathbf{y}$$

where

$$\mathbf{w}_y = \frac{n}{N} \Sigma_y^{-1} \mathbf{X}_{\text{pop}} (\mathbf{X}^T \Sigma_y^{-1} \mathbf{X} + \Sigma_\beta^{-1})^{-1} \mathbf{X}_{\text{pop}}^T \mathbf{N}$$

Here, the weights $\mathbf{w}_y$ depend on which y is being treated.

In a full Bayesian analysis, the variance parameters are unknown and assigned priors (usually of the minimally informative variety). See Gelman and Hill, *Data Analysis Using Regression and Multilevel/Hierarchical Models* (Cambridge, 2007), chaps. 16-18. Initial estimates can be obtained using the R `lmer` function, which treats the variance estimates as being known.

11.2. **Example.** Bayesian estimation of the H cell means model:

$$y_i \sim N(\beta_h, \sigma_y^2) \quad \text{if } i \in U_h$$

with the hierarchical prior

$$\beta_h \sim N(\mu, \sigma_\mu^2)$$

with a diffuse prior for β . The posterior means are

$$\hat{\beta}_h = \frac{(n_h/\sigma_y^2)\bar{y}_h + (1/\sigma_\beta^2)\hat{\mu}}{(n_h/\sigma_y^2) + (1/\sigma_\beta^2)}$$

$$\hat{\mu} = \frac{\sum_{h=1}^H \bar{y}_h/(\sigma_y^2/n_h + \sigma_\beta^2)}{\sum_{h=1}^H 1/(\sigma_y^2/n_h + \sigma_\beta^2)}$$

Gelman (p. 162) shows that the implied unit weights are for $k \in \omega_h$

$$w_k \frac{n}{\sigma_y^2 + n_h \sigma_\beta^2} \left(\frac{N_h \sigma_\beta^2}{N} + \frac{\sigma_y^2 \sum_{h=1}^H N_h/(\sigma_y^2 + n_h \sigma_\beta^2)}{N \sum_{h=1}^H n_h/(\sigma_y^2 + n_h \sigma_\beta^2)} \right)$$

Gelman observes that the ratio of sums is approximately N/n so the approximate weight is

$$w_k \approx \frac{n_h/\sigma_y^2}{n_h/\sigma_y^2 + 1/\sigma_\beta^2} \frac{N_h/N}{n_h/n} + \frac{1/\sigma_\beta^2}{n_h \sigma_y^2 + 1/\sigma_\beta^2}$$

which is a weighted average of the full poststratification weight and a weight of one.